PERSONALIZED FITNESS & DIET CHATBOT

Organization: Tata Consultancy Services (TCS) – Internship Project

Objective:

To build an AI-powered chatbot that generates personalized diet and workout plans based on user inputs like height, weight, age, goal, and lifestyle, and supports conversational interaction via WhatsApp.

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Key Features:

1. Diet Planning:

- Personalized meal plans.
- Calorie-based meal distribution.
- Protein optimization based on goal.

2. Calorie Calculation:

- BMR & TDEE-based calculations.
- Goal-based calorie adjustment.

3. Fitness Assistance:

- Workout suggestions.
- Goal-specific advice.

4. AI Chat System:

- Natural conversation using LLM (Ollama).
- Context-aware responses.
- Memory-based interactions.

5. Memory & Personalization:

- Stores chat history.
- Uses vector database (ChromaDB).
- Personalized experience per user.

6. UI Interaction:

- WhatsApp Bot (META API).

7. Smart Features:

- Edit profile capability.
- Plan modification.
- Guardrails (restricts non-fitness queries).

System Architecture:

Components Overview

1. Frontend Interface:

- WhatsApp API

2. Backend:

- FastAPI-based server

3. Core Engines:

- Calorie Engine
- Diet Engine
- Clinical Engine
- Workout Engine

4. AI Layer:

- Ollama (LLM: qwen2.5:7b)
- Embeddings (nomic-embed-text)

5. Database:

- PostgreSQL → structured data
- ChromaDB → vector memory

DATA FLOW:

- User sends message
- Request hits FastAPI webhook
- Input is validated & classified
- User profile fetched from DB
- Missing data collected via questions
- Once complete → plan generated
- Response sent back to user
- Data stored for future context

Core Modules (Detailed Explanation):

1. Calorie Engine:

This module calculates the **daily calorie requirement**.

Logic:

- **BMR Formula (Mifflin-St Jeor):**
  - Male:
    10W + 6.25H - 5A + 5
  - Female:
    10W + 6.25H - 5A – 161
- **TDEE Calculation:**
  - TDEE = BMR × Activity Factor
- **Goal Adjustment:**
  - Fat loss → calorie deficit
  - Muscle gain → calorie surplus

2. Diet Engine:

This is the core logic of your project.

Step-by-step logic:

1. **Dataset Loading:**
   - IFCT dataset used
   - Contains nutritional values
2. **Food Classification:**
   Each food is categorized into:
   - Protein-rich
   - Carb-rich
   - Other

3. **Meal Distribution:**

   Calories split into:

   - Breakfast → 25%
   - Lunch → 35%
   - Dinner → 25%
   - Snacks → 15%

4. **Smart Meal Selection:**
   Algorithm:
   - Prioritize foods based on meal type
   - Add items until calorie target is reached
   - Ensure protein intake optimization
5. **Output Formatting:**
   - Structured Readable meal plan

3. Clinical Engine:

Provides basic health recommendations.

Examples:

Hydration advice

Sleep recommendations

Sedentary lifestyle warnings

4. AI & Chat System:

Features:

- Natural language understanding
- Intent detection
- Context memory

Capabilities:

- Start plan
- Modify plan
- General chat
- Continue plan

Model:

- Ollama LLM (qwen2.5)

5. Guardrails:

Ensures chatbot remains domain-specific.

Logic:

- Checks if query is fitness-related
- Rejects unrelated queries

6. WhatsApp Integration:

Features:

- Text messages
- Button interactions
- List selection

Flow:

- Webhook receives message
- Processes user input
- Sends response via Meta API

DATABASE DESIGN

Tables:

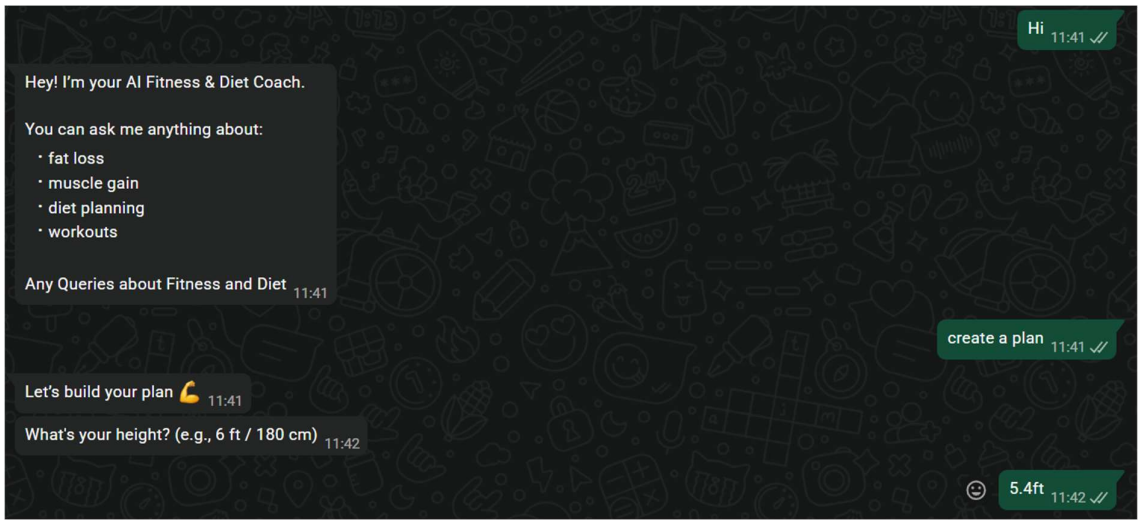
- user_profile
- conversation_context
- chat_memory
- plan_state
- processed_events
- conversation_artifacts

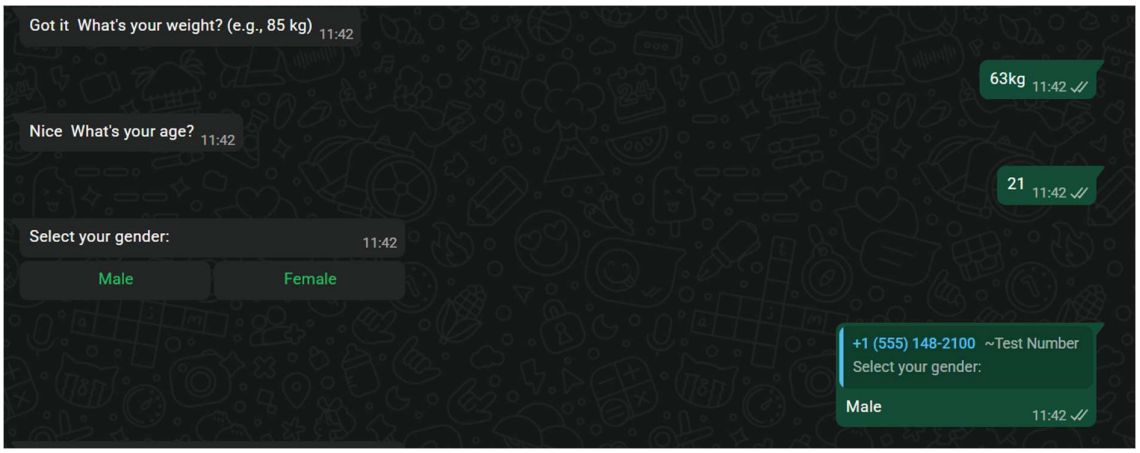
Purpose:

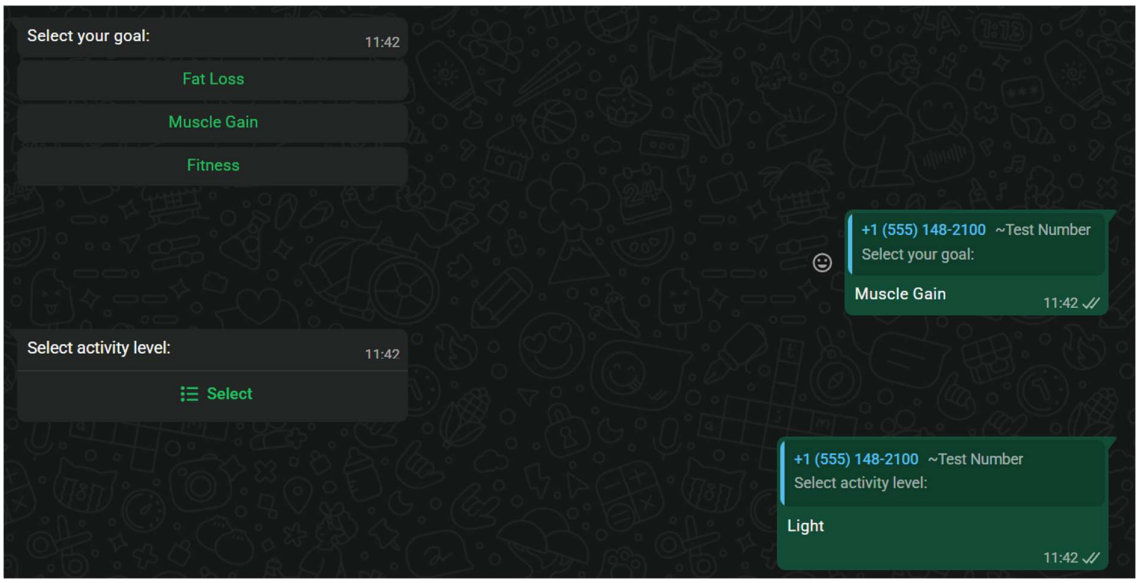
- Maintain user state
- Avoid duplicate processing
- Store conversation memory

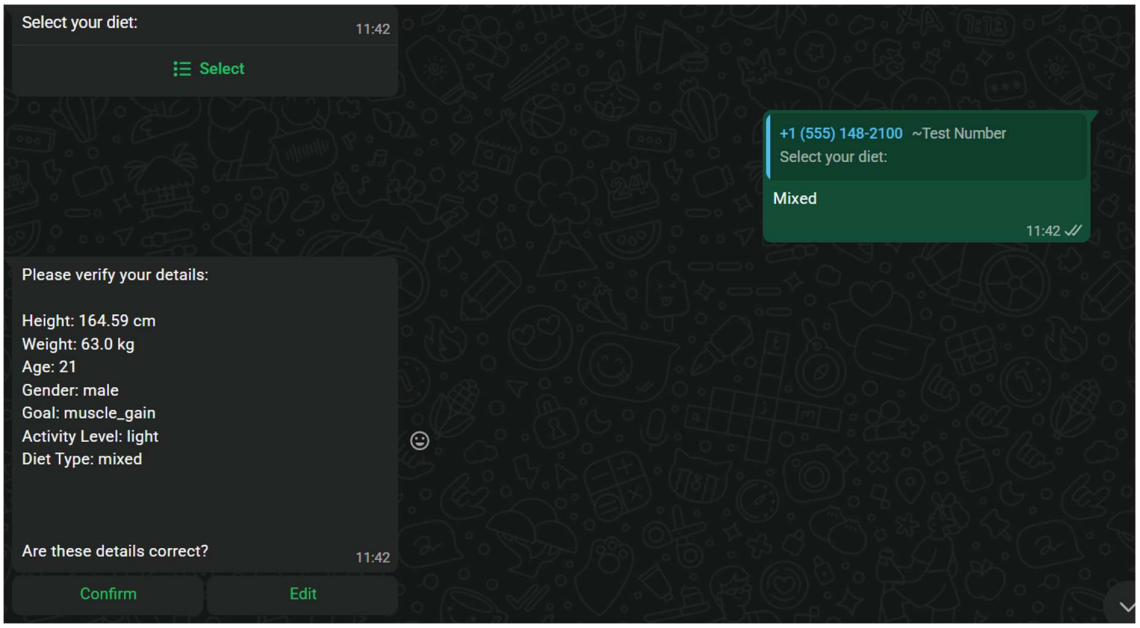
SCREENSHOTS

1. User Input flow (Questionnaire):

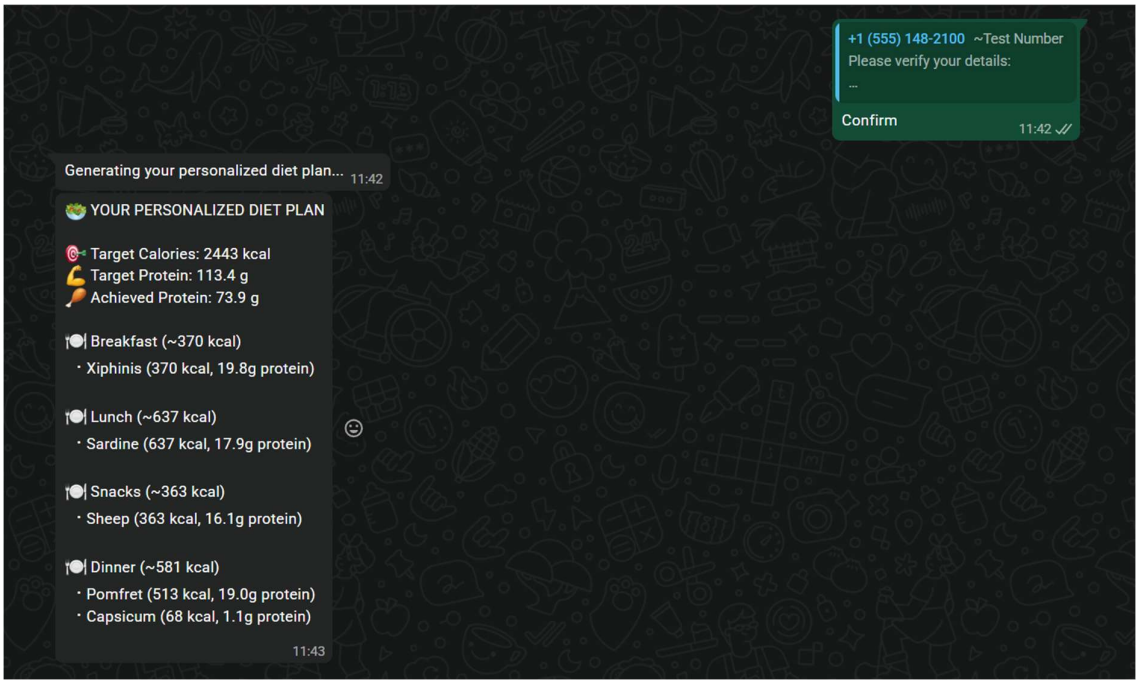



Got it What's your weight? (e.g., 85 kg) 11:42
63kg 11:42
Nice What's your age? 11:42
21 11:42
Select your gender: 11:42
Male
Female
+1 (555) 148-2100 ~Test Number
Select your gender:
Male 11:42

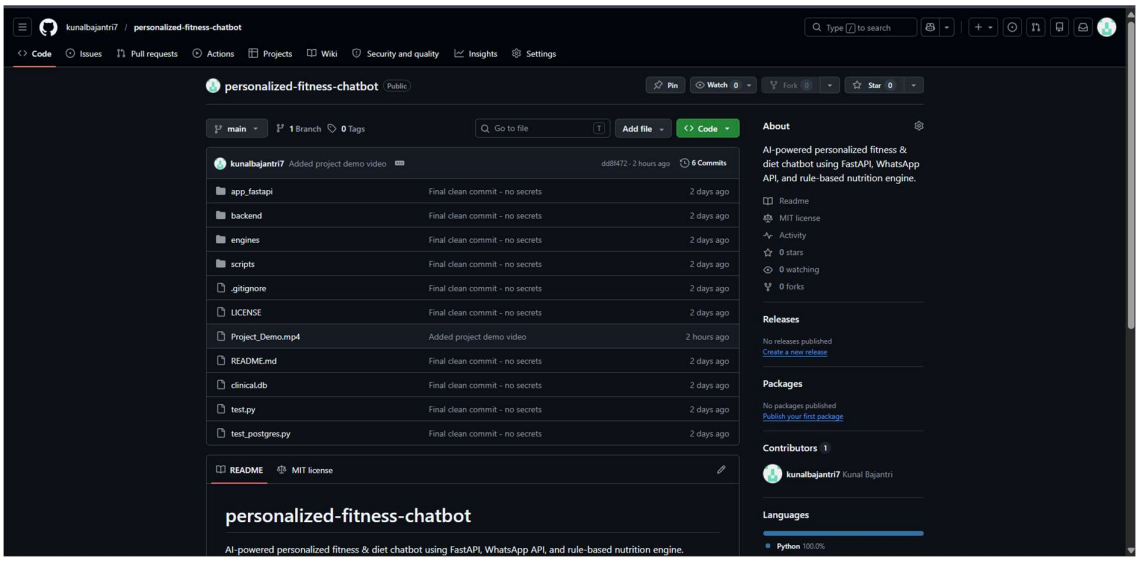

Select your goal: 11:42
Fat Loss
Muscle Gain
Fitness
+1 (555) 148-2100 ~Test Number
Select your goal:
Muscle Gain 11:42
Select activity level: 11:42
Select
+1 (555) 148-2100 ~Test Number
Select activity level:
Light 11:42


Select your diet: 11:42
Select
+1 (555) 148-2100 ~Test Number
Select your diet:
Mixed 11:42
Please verify your details:
Height: 164.59 cm
Weight: 63.0 kg
Age: 21
Gender: male
Goal: muscle_gain
Activity Level: light
Diet Type: mixed
Are these details correct? 11:42
Confirm
Edit

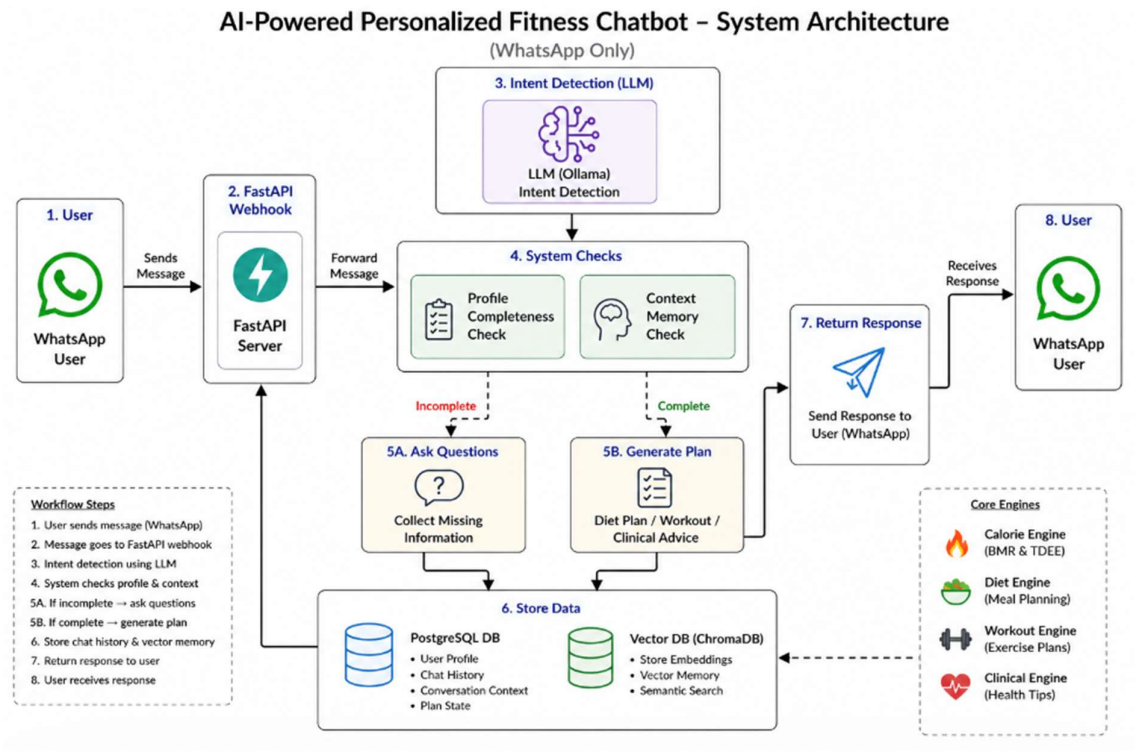
2. Diet Plan Output:



3. GitHub repository



4. Architecture Diagram



TESTING:

Tested Scenarios:

- New user onboarding
- Profile editing
- Diet generation
- Invalid input handling
- Non-fitness query rejection

DEPLOYMENT:

Run Backend:

uvicorn app_fastapi.main:app –reload

Webhook:

ngrok http 8000

Requirements:

- Python 3.10+
- PostgreSQL
- Ollama

GITHUB REPOSITORY:

https://github.com/kunalbajantri7/personalized-fitness-chatbot

Demo video is also present in the git repo.

CONCLUSION:

This project successfully demonstrates:

- AI-driven chatbot system
- Real-world backend architecture
- Personalized recommendation system
- WhatsApp UI integration

The system is scalable and can be extended into a full production-level fitness assistant.